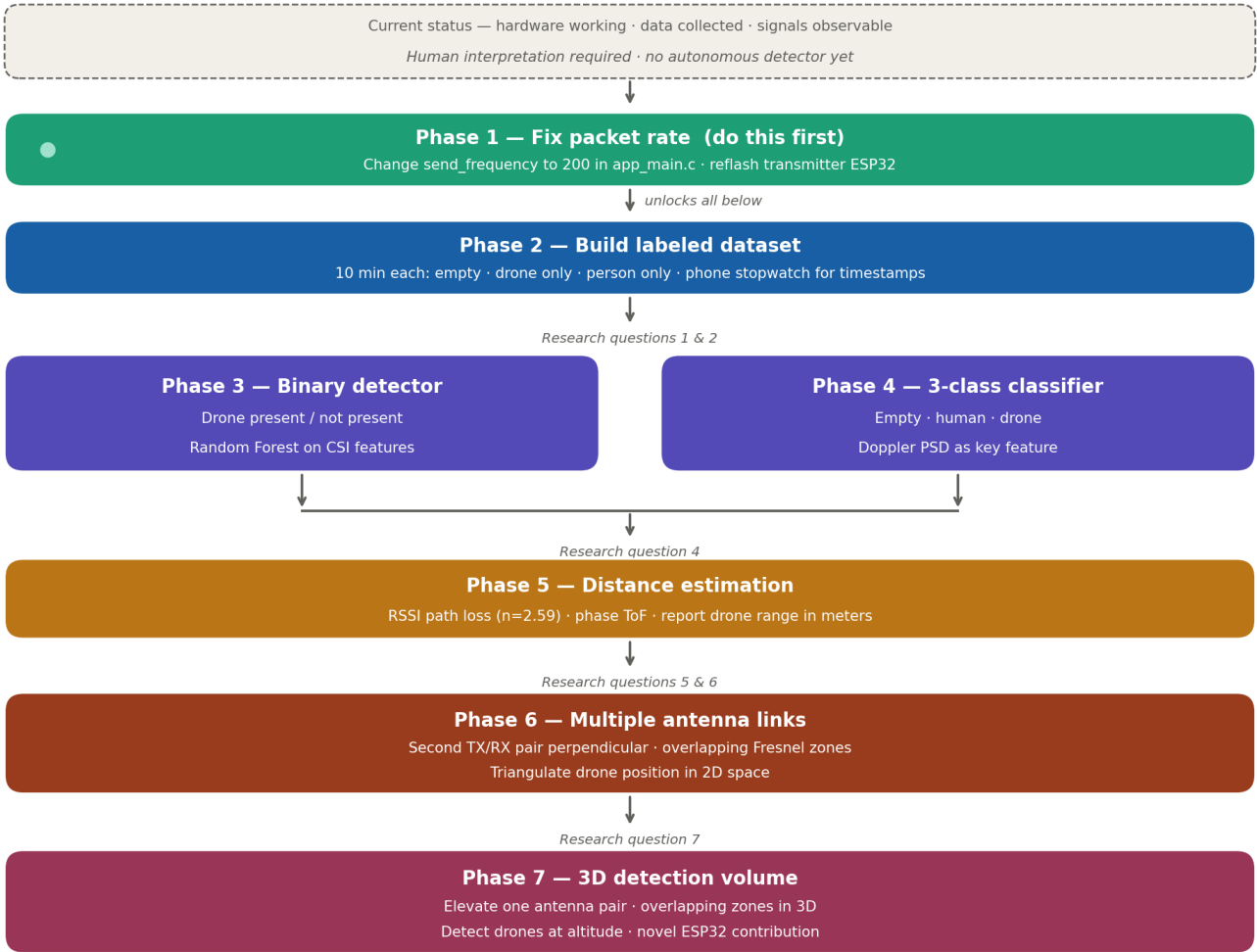


Drone Detection — Research Roadmap

ESP32 WiFi CSI System · University of New Orleans (UNO)



Minimum dataset per class (Phase 2)

Duration	Windows (2s, 50% overlap)	Sufficient for?
2 minutes	~120 windows	Too few — model will overfit
5 minutes	~300 windows	Borderline for Random Forest
10 minutes ✓	~600 windows	First working classifier
15 minutes	~900 windows	Good — adds generalization

Student Z · University of New Orleans (UNO) · March 2026

Base firmware: github.com/espressif/esp-csi

This roadmap represents the full research arc from the current experimental baseline to an autonomous 3D drone detection system. Phase 1 is the immediate priority — a single firmware parameter change that unlocks all subsequent phases. The dataset timing table (bottom) reflects the minimum collection needed for a first working classifier at 200 pps. Quality and accurate timestamps matter more than duration at this stage.

Student Z · University of New Orleans (UNO) · Professor P · March 2026 | Base firmware: github.com/espressif/esp-csi